

MET CS 555 – Foundation of Machine Learning

Module - 3

Lecture - 5: Linear Regression and Assessing the Fit

Lecture - 6: Multiple Linear Regression

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Simple Linear Regression (SLR)

If a linear relationship exists, **we can model** the nature of the relationship between the variables using **simple linear regression (SLR)**.

Using SLR, we assert a **straight line on the scatterplot** that represents the best fitting line to the data that captures the pattern of the relationship.

- ▷ An explanatory variable.
- ▷ A response variable

Simple Linear Regression (SLR)

We build a data model that allows us to:

- ▷ **quantify the relationship between the response variable and the explanatory variable**
- ▷ **predict the response of a new observation with a given value for x or what the average response is for observations with a specific value for the explanatory variable**

Simple Linear Regression (SLR)

The equation for the simple linear regression line is given by

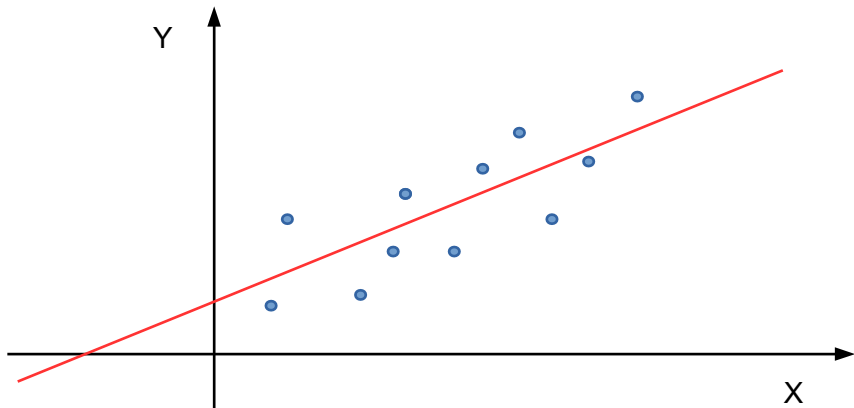
$$y = \beta_0 + \beta_1 x$$

- ▷ y is the response or dependent variable
- ▷ x is the explanatory or independent variable
- ▷ β_0 is the intercept (the value of y when $x = 0$)
- ▷ β_1 is the slope (the expected change in y for each one-unit change in x)

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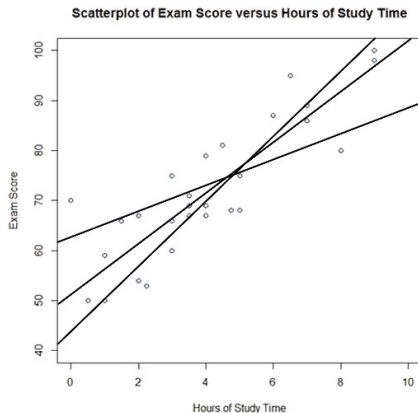
$$y = \beta_0 + \beta_1 x$$



How to find the regression line that best fits the data

There are several ways to find parameters of the line. The most common way is to **minimize the sum of the squares** of the distances between the points and the regression line.

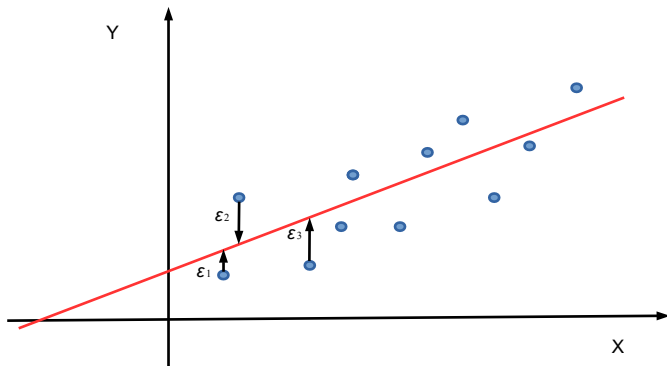
This approach is called the **least-squares method**. We want to minimize the vertical distance between each of the points and the regression line.



How to find the regression line that best fits the data

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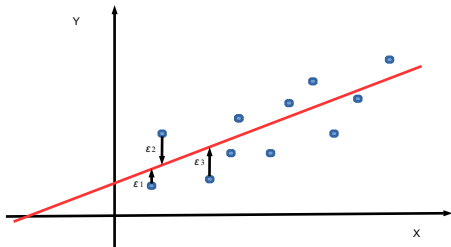
How to find the regression line that best fits the data

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

$$\epsilon_i = y_i - (\beta_0 + \beta_1 x_i)$$

$$f(\beta_0, \beta_1) = \sum_{i=1}^n \left(y_i - (\beta_0 + \beta_1 x_i) \right)^2$$

Loss function, l2 norm



How to find the regression line that best fits the data

$$y_i = \beta_0 + \beta_1 x_i + \epsilon$$

$$\text{Best Fit Slope: } \hat{\beta}_1 = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

$$\text{Best Fit Y-Intercept: } \hat{\beta}_0 = \frac{\sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i - \sum_{i=1}^n x_i \sum_{i=1}^n x_i y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

Equation for the least- squares regression line

The equation for the simple linear regression line is given by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

In the least-squares regression, the estimates of β_0 and β_1 are:

$$\hat{\beta}_1 = r \frac{s_y}{s_x}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

r is correlation coefficient

s_x the sample standard deviation of x , s_y is SD of y

$\bar{x}$ sample mean of x , and $\bar{y}$ sample mean of y

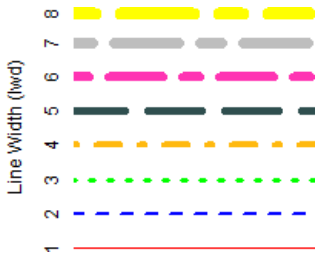
R Function - Simple linear regression

```
> lm(data$responsevariable ~ data$explanatory)
> abline(a=intercept, b=slope)

# create a model object
> m <- lm(score ~ study.hours)

# Add regression line to the scatterplot
> abline(m, lty=3, col="blue")
```

http://www.cookbook-r.com/Graphs/Shapes_and_line_types



An example - the least-squares regression line

National Unemployment Male vs. Female Reference: Statistical Abstract of the United States

```
> unemployment <- read.csv("national_unemployment_rate.csv")
> attach(unemployment)

> xbar <- mean(male.unemployment.rate)
> sx <- sd(male.unemployment.rate)
> ybar <- mean(female.unemployment.rate)
> sy <- sd(female.unemployment.rate)
> r <- cor(male.unemployment.rate, female.unemployment.rate)
> beta1 <- r*sy/sx
> beta0 <- ybar - beta1*xbar
```

$$\hat{\beta}_1 = 0.69$$

$$\hat{\beta}_0 = 1.43$$

$$\hat{y} = 1.43 + 0.69x$$

When $x = \bar{x} = 5.95$, $\hat{y} = \bar{y} = 5.57$

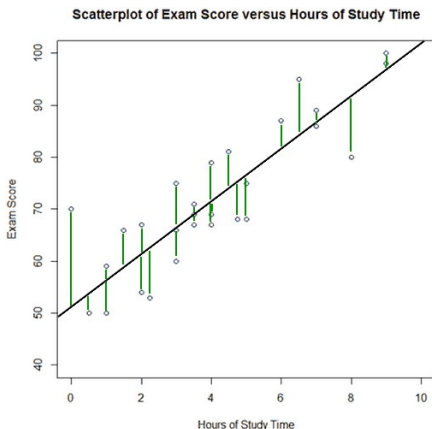
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How to find the regression line that best fits the data

There are several ways to find parameters of the line. The most common way is to **minimize the sum of the squares** of the distances between the points and the regression line.

This approach is called the **least-squares method**.



An Example Study Hours vs. Exam Scores

```
> student <- read.csv("student.csv")
> attach(student)

> xbar <- mean(study.hours)
> sx <- sd(study.hours)
> ybar <- mean(score)
> sy <- sd(score)
> r <- cor(study.hours, score)

> beta1 <- r*sy/sx
> beta1

> beta0 <- ybar - beta1*xbar
> beta0

#Use lm() function lm(data$responsevariable ~ data$explanatory)
> m <- lm(score ~ study.hours)
> summary(m)
```

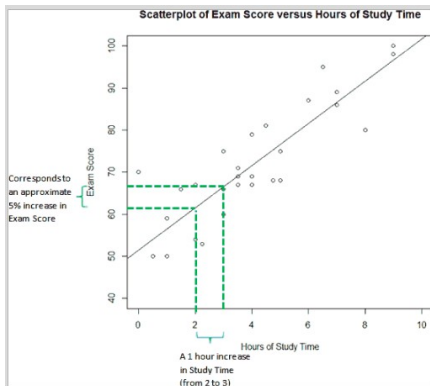
$$\hat{y} = 51.51 + 5.012x$$

Interpretation of results

Interpretation of results

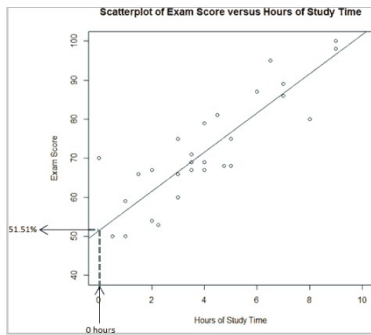
The estimate of the slope parameter ($\hat{\beta}_1$) gives the **expected or predicted change** in the response variable $\hat{y}$ **for a one-unit increase in the explanatory variable (x)**. Here, $\hat{\beta}_1 = 5.012$.

- ▷ Can be interpreted as the **increase in exam score for every one-hour increase in study time**.
- ▷ For **each additional hour** that students studied, their exam score **improved by around 5% points on average**.



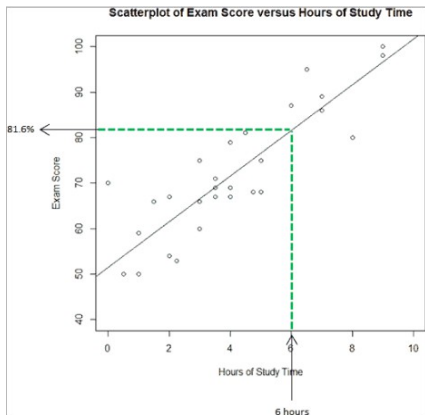
Interpretation of results

- ▷ The **Linear Nature** of the relationship and the equation implies that the increase in exam score is the same **for any 1 unit change**. **Increase from 3 to 2 hours is the same as increase from 9 hours versus 8 hours.**
- ▷ The estimate for the intercept (β_0) is meaningful in this case since values of the explanatory variable near 0 are possible. Here, $\beta_0 = 51.515$. **This can be interpreted as the average exam grade for those who did not study.**



Interpretation of Results - Example

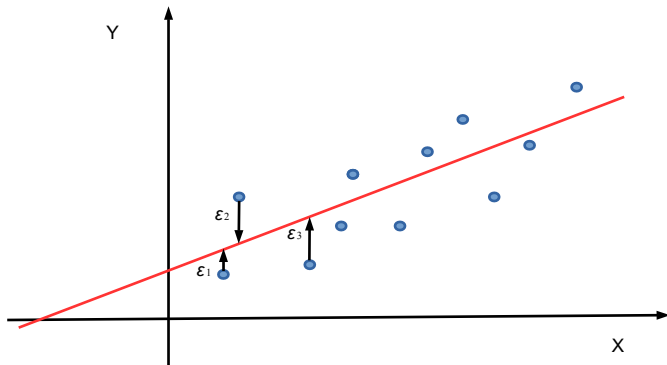
- ▷ The least-squares regression is $y = 51.51 + 5.012x$
- ▷ **If I were planning to study for 6 hours, what should I expect my score to be?**



- ▷ If I study for 6 hours by plugging in $x = 6$ into the regression equation $\hat{y} = 51.51 + 5.01x$
- ▷ That is, my average expected exam score is $y = 51.51 + 5.012(6) \approx 81.6$ if I study for 6 hours.

Random Error E

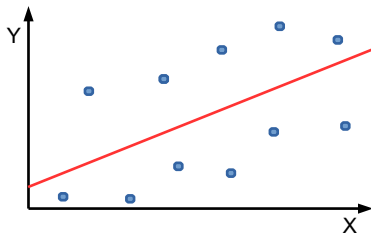
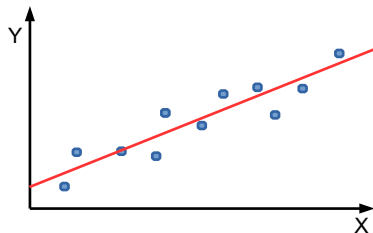
- ▷ The **true value of the response variable will vary** from the value predicted by the regression.
- ▷ We assume that **the random error term, E , is normally distributed** with a mean of 0 and a variance of σ^2 .
- ▷ The **larger the random error, the more the individual data points are scatter around** the linear regression line.
- ▷ We need to assess the **goodness of fit of the regression line**.



Assessing the Fit of Linear Regression Model

The coefficient of determination or R-squared

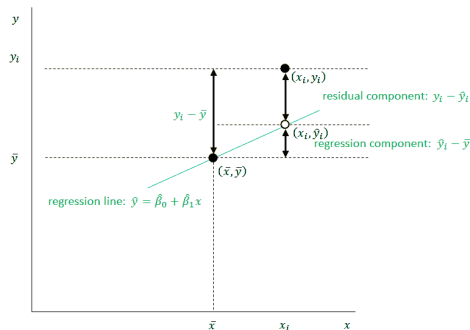
- ▷ The coefficient of determination, or R^2 , is a number that indicates how well data fit a statistical model - in our case, the regression line.
- ▷ R^2 represents the proportion (percentage) of the variation in the response variable **explained by the regression model** (equation).



Assessing the Fit of the Regression Line

For any given data point, the difference between the mean response and the observed response value y_i can be split into two parts:

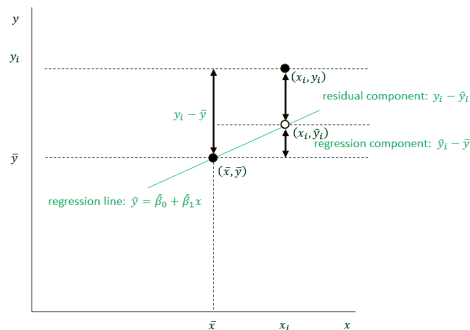
- (1) the residual component
- (2) the regression component



Assessing the Fit of the Regression Line

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For any sample point (x_i, y_i)

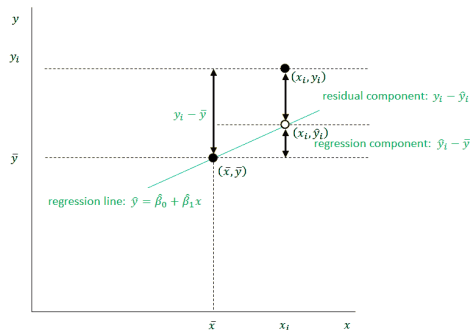
Residual component is the vertical distance between the **observed response**, y_i , and the regression **predicted response for x_i** .

Regression Component is the **vertical distance** between the regression **predicted response** for the value of explanatory variable x_i and the **average value of the response variable**.

Assessing the Fit of the Regression Line

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- (1) the residual component
- (2) the regression component



For any sample point (x_i, y_i)

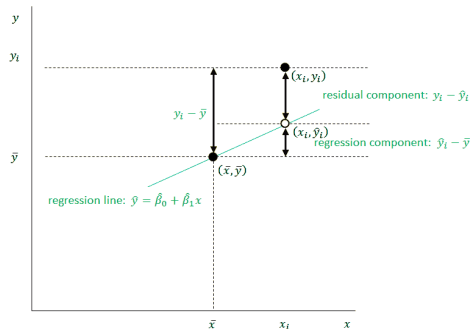
Residual component is $y_i - \hat{y} = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$

Regression Component is $\hat{y}_i - \bar{y} = (\hat{\beta}_0 + \hat{\beta}_1 x_i) - \bar{y}$

Assessing the Fit of the Regression Line

The **sum of the regression component and the residual component** gives us back the difference between the mean response $\bar{y}$ and the observed response value y_i

$$(y_i - \hat{y}_i) + (y_i - \bar{y}) = y_i - \hat{y}_i + y_i - \bar{y} = y_i - \bar{y}$$



- ▶ If **all data points fell on or very close to the regression line**, then $y_i \approx \hat{y}_i$ and the residual component $y_i - \hat{y}_i$ will be 0 or very close to 0.
- ▶ **A well-fitted regression line will have regression components that are larger than the residual components across all data points.**

The coefficient of determination or R-squared

Sum of all squared deviations and break it into each of the component parts to **see what proportion represents the regression components versus the residual components.**

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

where

- ▷ $\sum_{i=1}^n (y_i - \bar{y})^2$ (**Total Sum of Squares**) represents the sum of squares of the deviations of the individual sample points from the sample mean
- ▷ $\sum_{i=1}^n (y_i - \hat{y}_i)^2$ (**Residual Sum of Squares**) represents the sum of squares of the residual components
- ▷ $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ (**Regression Sum of Squares**) represents the sum of squares of the regression components

The Coefficient of Determination or R-squared

One of the measures that we use to **assess the fit of the data is the coefficient of variation** (R^2 , read "R-squared")

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\text{Reg SS}}{\text{Total SS}} = r^2$$

- ▷ **Coefficient of determination R^2** is the quantity that represents the proportion of the variation explained by the regression (or the "model").
- ▷ R^2 ranges between 0 and 1
- ▷ $R^2 = 1$ mean that model explains everything
- ▷ For SLR $R^2 = r^2$ Correlation Coefficient.

Why Assess the Linear Regression Model?

Why: To evaluate the overall performance, effectiveness, and predictive power of the model to check:

- ▷ **Accuracy:** Determine how well the model fits the data.
- ▷ **Validity:** Check if the assumptions underlying linear regression are met.
- ▷ **Reliability:** Ensure the model's predictions are consistent across different datasets.
- ▷ **Improvement:** Identify areas where the model can be refined.

How Assess the Linear Regression Model?

▷ **Model Fit:**

- **Coefficient of Determination (R^2):**

- . Measures how well the independent variables explain the variability in the dependent variable.

- . A higher R^2 indicates a better fit but not always, double check your data.

- **Adjusted R^2 :**

- . Adjusts R^2 for the number of predictors, helping to prevent overfitting.

▷ **Statistical Significance:**

- **P-values of Coefficients:**

- . Determine whether each predictor significantly contributes to the model.

- **Overall F-test:**

- . Tests whether the model provides a better fit than a model with no predictors.

Inference about Regression Coefficients

Inference about Regression Coefficients

- ▷ $\hat{\beta}_0$ and $\hat{\beta}_1$ are **statistics** calculated using sample data, not population parameters.
- ▷ If we **had a different sample**, we would get **different values of $\hat{\beta}_0$ and $\hat{\beta}_1$**
- ▷ Formal inference involves considering β_0, β_1 as unknown population parameters and determining what we can say about the unknown population parameters given the data we observed from our sample.

ANOVA and Linear Regression

- ▷ In linear regression, the primary goal is to explain as much of this variability in the dependent variable as possible using one or more independent variables (predictors).
- ▷ The Analysis of Variance (ANOVA) helps us assess whether the regression model significantly explains the variability in the dependent variable compared to what would be expected by chance.
- ▷ **what is variability in this context?** Variability, in this context, refers to the differences in the values of the dependent variable that you're trying to predict or explain using your regression model. It's a measure of how much the data points vary from one another and from the overall mean.
- ▷ **ANOVA and linear regression are related:**

Linear Regression: Models the relationship between dependent and independent variables.

ANOVA Table in Regression: Breaks down the variance explained by the model and the residual variance.

In regression, ANOVA is used to test the overall significance of the model.

Inference about Regression Coefficients

In regression analysis, assessment of the fit of the model to the data is performed using the quantities in the **ANOVA (analysis of variance) Table**.

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F-statistic	p-value
Regression	Reg SS	Reg df = k	Reg MS = Reg SS/Reg df	$F = \text{Reg MS} / \text{Res MS}$	$P(F_{\text{Reg df, Res df}, \alpha} > F)$
Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS/Res df		
Total	Total SS = Reg SS + Res SS				

- ▷ Total Sum of Squares (SST): measures the total variability in the dependent variable. The $df_{\text{Total}} = n - 1$ while n is the total number of observations. We lose one degree of freedom because we estimate the overall mean of the dependent variable. This represents the total number of independent deviations from the mean.
- ▷ Regression Sum of Squares (SSR): measures the variability explained by the regression model. The $df_{\text{Regression}} = k$ while k is the number of independent variables (predictors) in the model. Each predictor estimates one parameter (its coefficient), consuming one degree of freedom. Represents number of independent pieces of information used to estimate regression coefficients.
- ▷ Residual Sum of Squares (SSE): measures the variability not explained by the model (the errors or residuals). The $df_{\text{Residual}} = n - k - 1$. After estimating k coefficients for the predictors and the intercept (β_0), we lose $k + 1$ degrees of freedom. These are the degrees of freedom associated with the unexplained variability after fitting the model.

Inference about Regression Coefficients

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Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS / Res df		
Total	Total SS = Reg SS + Res SS				

Imagine the degrees of freedom as pieces of a pie:

▷ **Total Pie (df_total = n - 1):**

The entire variability in the dataset.

One slice is used to estimate the mean.

▷ **Regression Slice (df_reg = k):**

Portions of the pie allocated to the predictors.

Each predictor consumes one slice (degree of freedom) to estimate its coefficient.

▷ **Residual Slice (df_res = n - k - 1):**

The remaining pie after accounting for the mean and predictors.

Represents the variability unexplained by the model.

Inference about Regression Coefficients

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Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS/Res df		
Total	Total SS = Reg SS + Res SS				

- ▷ **Reg df = k** , the df of Reg SS, k = number of predictors in the model
- ▷ **Res df = $n - k - 1$** is the df of Res SS.
- ▷ **Reg MS = Reg SS/Reg df** (the regression mean square)
- ▷ **Res MS = Res SS/Res df** (the residual mean square)
- ▷ **F = Reg MS/Res MS** (F statistic value)
- ▷ **p-value** = The probability of observing a test statistic at least as extreme as the one calculated from your sample data, under the assumption that the null hypothesis is true.

Inference about Regression Coefficients

- ▷ **Reg MS = Reg SS/Reg df** (the regression mean square). It is a measure of the average variability explained by the regression model per degree of freedom.
- ▷ **Res MS = Res SS/Res df** (the residual mean square). measures the average variability in the dependent variable that is **not** explained by the regression model, per degree of freedom.
- ▷ **F = Reg MS/Res MS** (F statistic value). is a ratio used to compare the model's explained variance to the unexplained variance. It tests the overall significance of the regression model.

A larger F-statistic suggests that the model explains a significant portion of the variability in the dependent variable compared to the unexplained variability.

It is used in hypothesis testing to determine if at least one predictor variable is significantly related to the dependent variable.

- ▷ **p-value** = The **p-value** associated with the F-statistic represents the probability of observing an F-statistic as extreme as, or more extreme than, the one calculated from your data, assuming the null hypothesis is true.

- ▷ **Null Hypothesis (H_0)**: All regression coefficients (slopes) are equal to zero (no effect).
- ▷ **Alternative Hypothesis (H_a)**: At least one regression coefficient is not zero (at least one predictor has an effect).
- ▷ **Low p-value (typically ≤ 0.05)**: Suggests that the observed F-statistic is unlikely under the null hypothesis, leading us to reject H_0 and conclude that the model is significant.
- ▷ **High p-value (typically > 0.05)**: Suggests insufficient evidence to reject H_0 , the model may not be significant.

An example: Calculate R²

The association between husbands and wives ages was calculated to be $\hat{y} = -4.94 + 1.19x$. Using this and the fact that $\bar{y} = 26$, calculate by hand the quantities from the ANOVA table.

Calculate R-squared and give its interpretation.

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Calculate R-squared and give its interpretation.

Reg $df = 1$ for SLR, Res $df = n - k - 1 = n - 2 = 5 - 2 = 3$.

Couple	Age of Wife	Age of Husband
1	20	20
2	30	32
3	24	22
4	28	26
5	28	30
Sample mean	26	26
Sample standard deviation	4.0	5.1

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1	20	20
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3	24	22
4	28	26
5	28	30
Sample mean	26	26
Sample standard deviation	4.0	5.1

Couple	x_i	y_i	$\hat{y}_i$	$\hat{y}_i - \bar{y}$	$(\hat{y}_i - \bar{y})^2$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	20	20	18.86	-7.14	50.98	1.14	1.30
2	30	32	30.76	4.76	22.66	1.24	1.54
3	24	22	23.62	-2.38	5.66	-1.62	2.62
4	28	26	28.38	2.38	5.66	-2.38	5.66
5	28	30	28.38	2.38	5.66	1.62	2.62
Sum					90.63		13.75

An example: Calculate R^2

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)
Regression	Reg SS = 90.63	Reg df = $k = 1$	Reg MS = $90.63/1 = 90.63$
Residual	Res SS = 13.75	Res df = $n - k - 1 = 5 - 1 - 1 = 3$	Res MS = $13.75/3 = 4.58$
Total	Total SS = Reg SS + Res SS = 104.38		

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\text{Reg SS}}{\text{Total SS}} = \frac{90.63}{104.38} = 86.8\%$$

86.8% of the variability in husbands ages can be explained by wives' ages.

Formal Tests of Hypotheses

In SLR, formal tests of hypotheses concern β_1 .

They are generally of the form

$H_0 : \beta_1 = 0$ (there is no linear association)

$H_1 : \beta_1 \neq 0$ (there is a linear association)

- ▷ $\beta_1 = 0$ would mean that the regression line had a slope of 0 (**a horizontal line**)
- ▷ $\beta_1 = 0$ is equivalent to saying that the **response variable does not change at all when the explanatory variable changes.**

F-test for Simple Linear Regression

$H_0 : \beta_1 = 0$ (there is no linear association)

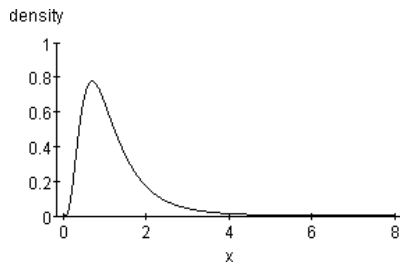
$H_1 : \beta_1 \neq 0$ (there is a linear association)

$$F = \frac{\text{Reg MS}}{\text{Res MS}}$$

F-distribution with 1 and $n - 2$ degrees of freedom under H_0 .

F-distribution

- ▷ The **F-distribution** is named after the famous statistician R. A. Fisher¹.
- ▷ **F is the ratio of two variances.**
- ▷ The F-distribution is most commonly used in **Analysis of Variance (ANOVA) and the F test** (to determine if two variances are equal).
- ▷ It has a **minimum of 0, but no maximum value** (all values are positive).
- ▷ The peak is not far from 0.
- ▷ When referencing the F-distribution the numerator degrees of freedom are always given first, and switching the degrees of freedom changes the distribution (**$F(10,12)$ does not equal $F(12,10)$**).



¹https://en.wikipedia.org/wiki/Ronald_Fisher

F-distribution

Why Does the F-Distribution Have Two Degrees of Freedom?

The F-distribution has two degrees of freedom because it is derived from the ratio of two independent chi-squared variables, each with its own degrees of freedom.

Components:

•Numerator Degrees of Freedom:

- Associated with the variance estimate in the numerator of the F-statistic.
- In regression, this often corresponds to the number of predictors (or groups in ANOVA) minus one.

•Denominator Degrees of Freedom:

- Associated with the variance estimate in the denominator of the F-statistic.
- In regression, this is typically the residual degrees of freedom (total observations minus the number of estimated parameters).

Why Is the Peak of the F-Distribution So Sharp?

The sharpness of the peak near zero in the F-distribution is influenced by the degrees of freedom associated with the numerator and denominator variances.

Factors Contributing to a Sharp Peak:

Degrees of Freedom (df):

Small Numerator Degrees of Freedom:

When it is small, the distribution has a sharper peak near zero.

Large Denominator Degrees of Freedom:

As it increases, the distribution becomes more concentrated around lower F-values.

Variability of Estimates:

With fewer degrees of freedom, variance estimates are less reliable and more variable, leading to a higher probability of small F-values.

F-distribution

Why Does the F-Distribution Have Two Degrees of Freedom?

The F-distribution has two degrees of freedom because it is derived from the ratio of two independent chi-squared variables, each with its own degrees of freedom.

In simple linear regression:

Numerator degrees of freedom ($=1$):

Corresponds to the number of predictors (one in simple regression).

Denominator Degrees of Freedom ($= n-2$):

Total observations minus the number of estimated parameters (intercept and slope).

F-test for Simple Linear Regression

$H_0 : \beta_1 = 0$ (there is no linear association)

$H_1 : \beta_1 \neq 0$ (there is a linear association)

$$F = \frac{\text{Reg MS}}{\text{Res MS}}$$

F-distribution with 1 and $n - 2$ degrees of freedom under H_0 .

The decision rule is:

Reject $H_0 : \beta_1 = 0$ if $F \geq F_{1,n-2,\alpha}$

Otherwise, do not reject $H_0 : \beta_1 = 0$

where **$F_{1,n-2,\alpha}$ is the value from the F-distribution** with 1 degree of freedom (numerator) and $n - 2$ degrees of freedom (denominator) and associated with a right-hand tail probability of α .

R Functions - Quantities from the F-distribution

```
# Calculating probability from F-statistics
# Use pf() function to calculate the area to the left of a given F-
  statistic
> pf([F statistic], df1=[degree of freedom of the numerator], df2=[
  degree of freedom of the denominator])

# Calculating F-statistics from probability
# Use qf() function to calculate F-statistic with the specifies area to
  the left
> qf([probability], df1=[degree of freedom of the numerator], df2=[
  degree of freedom of the denominator])
```

Quantities from the F-distribution

```
> pf(18.51, df1=1, df2=2)
1 0.9499929 # (the area to the left)
> pf(18.51, df1=1, df2=2, lower.tail = F) # (area to the right)
[1] 0.05000706
> qf(0.95, df1=1, df2=2) # getting the p-value out of F-value
[1] 18.51282
```

Table C. *F*-Distribution Critical Values

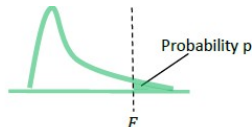


Table entry for p is the critical value F with probability p lying to its right

		Degrees of freedom in the numerator									
	p	1	2	3	4	5	6	7	8	9	10
1	0.100	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86	60.19
	0.050	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
	0.025	647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28	968.63
	0.010	4052.18	4999.50	5403.35	5624.58	5763.65	5858.99	5928.36	5981.07	6022.47	6055.85
	0.001	405284	499999	540379	562500	576405	585937	592873	598144	602284	605621
2	0.100	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39
	0.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
	0.025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40
	0.010	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40
	0.001	998.50	999.00	999.17	999.25	999.30	999.33	999.36	999.37	999.39	999.40
	0.100	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23

A Example: F-test for Simple Linear Regression

Is there a linear relationship between hours of study time and exam score? Perform this test at the $\alpha = 0.05$ level.

1. Set up the hypotheses and select the alpha level

$H_0 : \beta_1 = 0$ (there is no linear association)

$H_1 : \beta_1 \neq 0$ (there is a linear association)

$\alpha = 0.05$

2. Select the appropriate test statistic

$$F = \frac{MS \text{ Reg}}{MS \text{ Res}}$$

with 1 and $n-2 = 31-2 = 29$ degrees of freedom

A Example: F-test for Simple Linear Regression (Continued)

3. State the decision rule

F-distribution with 1, 29 degrees of freedom and associated with $\alpha = 0.05$.

```
> qf(.95, df1=1, df2=29)
```

$$F_{1,29,0.05} = 4.1830$$

Decision Rule: Reject H_0 if $F \geq 4.1830$

Otherwise, do not reject H_0

A Example: F-test for Simple Linear Regression (Continued)

4. Compute the test statistic

Using R function `anova()`, we got the following ANOVA table:

	SS	df	MS	F-statistic	p-value
Regression	4973.5	1	4973.5	103.2	4.625e-11
Residual	1398.0	29	48.2		
Total					

$$F = \frac{MS_{\text{Reg}}}{MS_{\text{Res}}} = \frac{4973.5}{48.2} \approx 103.2 \text{ with 1 and 29 degrees of freedom.}$$

F-statistic can also be calculated using `summary()`

5. Conclusion

- ▷ Reject H_0 since $103.2 \geq 4.183$.
- ▷ We have significant evidence at the $\alpha = 0.05$ level that $\beta_1 \neq 0$.
- ▷ There is evidence of a significant linear association between study time and exam score (here, $p < 0.001$ as calculated using software program).

Inference from regression - t-test

In linear regression, the sampling distribution of the coefficient estimates form a normal distribution, which is approximated by a **t distribution due to approximating σ by s** .

We can calculate a confidence interval for each estimated coefficient or **perform a hypothesis test**:

$H_0 : \beta_1 = 0$ (there is no linear association)

$H_1 : \beta_1 \neq 0$ (there is a linear association)

Test Statistic **t-test**

$$t = \frac{\hat{\beta}_1}{SE_{\hat{\beta}_1}}$$

with $df = n - 2 = 31 - 2 = 29$ degrees of freedom

Normality Assumption - Standard Error of Regression Coefficients

How is the standard error of the estimator $\hat{\beta}$ calculated?

$$s_{\hat{\beta}} = \sqrt{\frac{\frac{1}{n-2} \sum_{i=1}^n \hat{\varepsilon}_i^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

No worries! We use R or Python to compute this.

T-test for SLR - Confidence Interval

The decision rule for a two-sided level α t-test is:

Reject $H_0 : \beta_1 = 0$ if $|t| \geq t_{n-2, \frac{\alpha}{2}}$ OR $p \leq \alpha$

Otherwise, do not reject $H_0 : \beta_1 = 0$

where $t_{n-2, \frac{\alpha}{2}}$ is the value from the t-distribution table with $n - 2$ degrees of freedom and associated with a **right hand tail probability of $\alpha/2$** .

The two-sided $100\% \times (1 - \alpha)$ confidence interval for β_1 is given by:

$$\hat{\beta}_1 \pm t_{n-2, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_1}$$

Interpretation: We can say with 95% confidence (for standard $\alpha = 0.05$) that the true value of β_1 is between

$$\hat{\beta}_1 - t_{n-2, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_1} \text{ and } \hat{\beta}_1 + t_{n-2, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_1}$$

A Example: t-test for Simple Linear Regression

Is there a linear relationship between hours of study time and exam score?

Perform a t-test at the $\alpha = 0.05$ level, construct and interpret the 95% confidence interval for β_1 .

1. Set up the hypotheses and select the alpha level

$H_0 : \beta_1 = 0$ (there is no linear association) $H_1 : \beta_1 \neq 0$ (there is a linear association) $\alpha = 0.05$

2. Select the appropriate test statistic

$$t = \frac{\hat{\beta}_1}{SE_{\hat{\beta}_1}} \quad (1)$$

with $df = n - 2 = 31 - 2 = 29$ degrees of freedom

3. State the decision rule Determine the appropriate value from the t-distribution table with 29 degrees of freedom and associated with a right hand tail probability of $\alpha/2 = 0.025$

A Example: t-test for Simple Linear Regression (Continued)

3. State the decision rule Determine t-value with $df = 29$ and associated with a right hand tail probability of $\alpha/2 = 0.025$

$$t_{n2,\alpha/22} = t_{29,0.025} = 2.045$$

Decision Rule: Reject H_0 if $t \geq 2.045$ or if $t \leq -2.045$

Otherwise, do not reject H_0

4. Compute the test statistic. Using R we can have

	Estimate	SE	t-statistic	p-value
Intercept	51.5147	2.3820	21.63	2e-16
Hours	5.0121	0.4934	10.16	4.63e-11

$$t = \frac{\hat{\beta}_1}{SE_{\hat{\beta}_1}} = \frac{5.0121}{0.4934} \approx 10.16$$

A Example: t-test for Simple Linear Regression (Continued)

Also we can calculate the confidence interval

$$\begin{aligned}\hat{\beta}_1 \pm t_{n-2, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_1} &= 5.0121 \pm 2.045 \cdot 0.4934 \\ &= (4.00, 6.02)\end{aligned}$$

5. Conclusion

- ▷ Reject H_0 since $10.16 \geq 2.045$.
- ▷ We have significant evidence at the $\alpha = 0.05$ level that $\beta_1 \neq 0$. That is, there is evidence of a significant linear association between study time and exam score among students in CS546 (here, $p < 0.001$ as calculated using software program).
- ▷ We are 95% confident that the true value of is between 4.00 and 6.02.

R Functions - lm(), anova(), fitted(), resid()

```
# create linear model object
my.model <- lm( student$score ~ student$hours)

# report a summary of the model, reports regression coefficients,
  RSquared, F values, t values, p-values
summary( my.model)

# create Analysis of Variance Table
anova( my.model)

# fitted values of my model
fitted( my.model)

# residuals of my model
resid( my.model)
```

Adjusted R^2

R Squared is the estimate of the variability of the response variable y given a particular value of the explanatory variable x . It is a statistic and depends on two parameters, 1. number of samples, 2. number of variables in the model (in SLR $k=1$).

We want to be more conservative and adjust (reduce) it to state claims that are more true.

$$R_{adj}^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - k - 1} \right)$$

- ▷ n is the number of points in your data sample.
- ▷ k is the number of independent regressors, i.e., the number of variables in the model (in SLR $k=1$)
- ▷ Always $R_{adj}^2 \leq R^2$
- ▷ In SLR for large n , $R_{adj}^2 \approx R^2$

Closing Remarks - Some key points

- ▷ **Correlation between x and y is independent of order**
- ▷ Regression and correlation will give same conclusion, **but regression coefficient depends on which variable is specifies as explanatory**
- ▷ **In regression, t-test and F-test give same result - same p-value**
- ▷ **In SLR, $F = t^2$**
- ▷ **t-test from correlation and t-test from regression are equivalent**
and also give same result

MET CS 555 – Foundation of Machine Learning

Module - 3

Lecture -6: Multiple Linear Regression

Tahereh Javaheri

Boston University

Table of contents

1. [Multiple Linear Regression](#)

Multiple Linear Regression

Multiple Linear Regression (MLR)

- ▷ MLR can be used to describe the relationships between a set of explanatory or independent variables $(x_1, x_2, \dots, x_k)$ and a dependent variable (y)
- ▷ We are interested in the relationship between **each independent variable** and the dependent variable **after accounting for remaining independent variables**.
- ▷ MLR allows **quantifying the relationship** between our response variable and our explanatory variables as well as providing a tool for **predicting the response of a new observation** for a given set of values for $x_1, x_2, \dots$, and x_k .

Multiple Linear Regression (MLR)

The equation for the simple linear regression line is given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + e$$

- ▷ y is the response or dependent variable
- ▷ $x_1, x_2, \dots, x_k$ are the explanatory or independent variables
- ▷ β_0 is the intercept (the value of y when the $x_1, x_2, \dots, x_k$ are set to 0)
- ▷ β_1 is the slope (the expected change in y for each one-unit change in x_1 after adjusting for $x_2, \dots, x_k$)
- ▷ β_k is the slope (the expected change in y for each one-unit change in x_k after adjusting for $x_1, x_2, \dots, x_{k-1}$)
- ▷ e is the random error which we assume is normally distributed with a mean of 0 and a variance of σ^2

Regression Equation

The equation for the multiple linear regression line is given by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k$$

where

- ▷ $\hat{y}$ (read y hat) is the expected or predicted value of y for a given values of $x_1, x_2, \dots, x_k$
- ▷ $\hat{\beta}_0$ is the least-squares estimates of (the intercept)
- ▷ $\hat{\beta}_1, \hat{\beta}_2, \dots$ and $\hat{\beta}_k$ are the least-squares estimates of $\beta_1, \beta_2, \dots$ and β_k respectively

Regression Equation

The equation for the multiple linear regression line is given by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k$$

In the least-squares regression, the estimates are selected in such a way that the following quantity is **minimized**:

$$(y - \hat{y})^2 = y - \hat{\beta}_0 - \hat{\beta}_1 x_1 - \hat{\beta}_2 x_2 - \dots - \hat{\beta}_k x_k \quad ^2$$

A MLR Example

In the book Blink by Malcolm Gladwell, Gladwell states that a study of CEOs of Fortune 500 companies found that these individuals tend to be taller than the average US population.

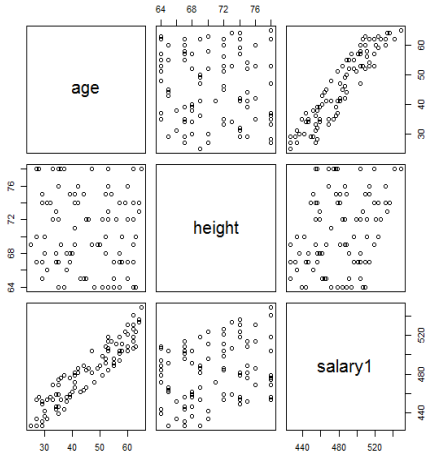
In order to study this phenomenon in more detail and to see if height is associated with increased success in business (as measured by salary), 100 men between the ages of 25 and 65 were polled for their heights (in inches) and annual salaries.

A MLR Example R Commands - Scatterplot Matrix

```
# Scatterplot Matrix
> data <- read.csv("CEO_salary.csv")
> attach(data)
> salary1 <- salary/1000
> data1 <- data.frame(age, height, salary1)
> cor(data1)
> pairs(data1)
```

A MLR Example R Commands - Scatterplot Matrix

```
# Scatterplot Matrix  
> data <- read.csv("CEO_salary.csv")  
> attach(data)  
> salary1 <- salary/1000  
> data1 <- data.frame(age, height, salary1)  
> cor(data1)  
> pairs(data1)
```



R Command Function

Finding the regression coefficients.

```
# Build a multiple linear regression model.
```

```
> m <- lm(salary1 ~ age + height)
```

```
> m
```

Call:

```
lm(formula = salary1 ~ age + height)
```

Coefficients:

(Intercept)	age	height
190.697	2.503	2.507

```
# pass the model to summary function
```

```
> summary(m)
```

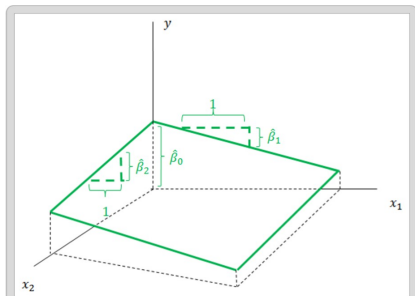
Interpretation of a regression equation

- ▷ **Focus is more on slope parameters**
- ▷ The slope parameter ($\hat{\beta}_i$) in a MLR gives the expected or average change in the response variable (y) for a one unit increase in the independent variable (x_i) after controlling for the other independent variables.
- ▷ **Beta estimates from MLR regression is not the same as beta for separate SLR**

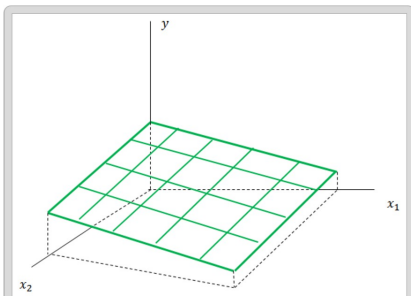
We use the equation of the least-squares regression line **to predict the expected value of the response variable for new values of the explanatory variables.**

Interpretation of the regression line

In MLR, the regression line corresponds to a plane ($k = 2$) or a hyperplane ($k > 2$, a k -dimensional plane in a $k+1$ -dimensional space).



The beta estimates define the surface of the plane. $\hat{\beta}_0$ is the expected value of y when x_1 and x_2 are 0. $\hat{\beta}_1$ is the slope of the surface projected onto the x_1 , y plane. $\hat{\beta}_2$ is the slope of the surface projected onto the x_2 , y plane.



The three dimensional plane is highlighted here. The perpendicular lines on the plane help reinforce the fact that for any given value of x_1 (for example), the regression asserts the same straight line relationship between x_2 and y . Similarly, for any given value of x_2 (for example), the regression asserts the same straight line relationship between x_1 and y .

Assessing the Fit of the Regression Line - coefficient of determination

The coefficient of determination represents the proportion (percentage) of the variation in the response variable explained by the multiple regression model.

- ▷ Coefficient of determination (R^2) is the same as SLR - in MLR we have more variables.
- ▷ Given that there are more than one independent variable in this setting, the **coefficient of determination is not simply the squared correlation coefficient.**

In MLR, R^2 is referred to as the multiple R-squared.

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\text{Reg SS}}{\text{Total SS}}$$

Inference Global F test

- ▷ We use the ANOVA table and are interested in testing the **alternative hypothesis that at least one of the slope parameters is different than 0.**
- ▷ If this test confirms that there is at least one slope parameter that is different than 0, then **subsequent F-tests or t-tests** can be used to assess whether each individual slope parameter is different than 0.
- ▷ In MLR, the F-test for the model is referred to as the **global test.**

Inference Global F test

- ▷ We use the ANOVA table and are interested in testing the **alternative hypothesis that at least one of the slope parameters is different than 0**.
- ▷ If this test confirms that there is at least one slope parameter that is different than 0, then **subsequent F-tests or t-tests** can be used to assess whether each individual slope parameter is different than 0.
- ▷ In MLR, the F-test for the model is referred to as the **global test**.

Difference to SLR is that, here **$k > 1$** . The exact value of k depends on the number of variables in the model.

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F-statistic	p-value
Regression	Reg SS	Reg df = k	Reg MS = Reg SS / Reg df	$F = \text{Reg MS} / \text{Res MS}$	$P(F_{\text{Reg df, Res df}, \alpha} > F)$
Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS / Res df		
Total	Total SS = Reg SS + Res SS				

Inference - Anova Table Components

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F-statistic	p-value
Regression	Reg SS	Reg df = k	Reg MS = Reg SS/Reg df	$F = \text{Reg MS}/\text{Res MS}$	$P(F_{\text{Reg df, Res df}, \alpha} > F)$
Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS/Res df		
Total	Total SS = Reg SS + Res SS				

- ▷ $\text{Reg SS} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$, $\text{Res SS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$, $\text{Total SS} = \sum_{i=1}^n (y_i - \bar{y})^2$
- ▷ **Reg df = k equals to the number of predictors in the model.**
- ▷ **Res df = n - k - 1** equals to sample size minus the number of predictors in the model minus 1.
- ▷ $\text{Reg MS} = \text{Reg SS}/\text{Reg df}$ (the regression mean square)
- ▷ $\text{Res MS} = \text{Res SS}/\text{Res df}$ (the residual mean square)
- ▷ $F = \text{Reg MS}/\text{Res MS}$
- ▷ p-value = the probability that the observed value of test statistic or a more extreme value could have been observed by chance

An Example: Calculate R^2

As displayed in the scatterplot matrix, CEOs salary is strongly associated with their age, and somewhat associated with their height.

The least-squares regression equation for the data looking at the association between CEOs salary and these factors was calculated to be

$$\hat{y} = 191 + 2.5 * age + 2.5 * height.$$

Calculate the multiple R-squared and give its interpretation.

$$\text{Reg df} = 2 \text{ and Res df} = n - k - 1 = n - 3 = 100 - 3 = 97.$$

An Example: Calculate R^2 (continued)

```
> totalss <- sum((salary1 - mean(salary1))^2)
> regss <- sum((fitted(m) - mean(salary1))^2)
> resiss <- sum((salary1 - fitted(m))^2)
> fstatistic <- (regss/2)/(resiss/97)
> pvalue <- 1-pf(fstatistic, df1=2, df2=97)
> R2 <- regss/totalss
```

99% of the variability in CEOs salary can be explained by age and height.

Global F-Test

In MLR, the first formal tests of hypotheses is for the overall model.

They test the null hypothesis that

- ▷ all slope coefficients are equal to 0

$$(H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0)$$

The null hypothesis is the same as asserting that there is **no linear relationship between the response and explanatory variables.**

- ▷ alternative that at least one of the slope coefficients is different from zero
 $(H_1 : \beta_i \neq 0 \text{ for at least one } i).$

The null hypothesis is rejected if there is at least one that is sufficiently far from 0 or (equivalently) a large majority of the total sum of squares is explained by the regression.

F-Test for MLR

$$F = \frac{\text{MS Reg}}{\text{MS Res}}$$

F-distribution with k and $n - k - 1$ degrees of freedom under H_0 .

The decision rule for a level α test is:

Reject $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ if $F \geq F_{k, n-k-1, \alpha}$

Otherwise, do not reject H_0

where **$F_{k, n-k-1, \alpha}$ is the value from the F-distribution** with

- ▷ k degree of freedom (numerator) and
- ▷ $n - k - 1$ degrees of freedom (denominator) and
- ▷ associated with a right-hand tail probability of α .

Quantities from the F-distribution

```
# Calculating probability from F-statistics
# Use pf() function to calculate the area to the left of a given F-
  statistic

> pf([F statistic], df1=[degree of freedom of the numerator], df2=[
  degree of freedom of the denominator])

# Calculating F-statistics from probability
# Use qf() function to calculate F-statistic with the specifies area to
  the left

> qf([probability], df1=[degree of freedom of the numerator], df2=[
  degree of freedom of the denominator])
```

A Example: F-test for MLR

Is there a linear relationship between COE's salary and their age and height? Perform this test at the $\alpha=0.01$ level.

1. Set up the hypotheses and select the alpha level

$H_0 : \beta_{age} = \beta_{height} = 0$ (age and height are not significant predictors of annual salary)

$H_1 : \beta_{age} \neq 0$ and/or $\beta_{height} \neq 0$ (at least one of the slope coefficients is different than 0; age and/or height are significant predictors/is a significant predictor of annual salary))

$$\alpha = 0.01$$

2. Select the appropriate test statistic

$$F = \frac{MS \text{ Reg}}{MS \text{ Res}}$$

with 2 and $n-3=97$ degrees of freedom

An Example: F-test for MLR

3. State the decision rule

F-distribution with 2, 97 degrees of freedom and associated with $\alpha = 0.01$.

```
> qf(.99, df 1 = 2, df 2 = 97)
```

$$F_{2,97,0.01} = 4.83$$

Decision Rule: Reject H_0 if $F \geq 4.83$

Otherwise, do not reject H_0

An Example: F-test for MLR

4. Compute the test statistic

```
> fstatistic <- (regss/2)/(resiss/97)
3.56445 e+10

# Or
# pass the model reference to summary function
> summary(m)

# Or
> pvalue <- 1-pf(fstatistic, df1=2, df2=97)
```

5. Conclusion

Reject H_0 since f statistic ≥ 4.83 .

We have significant evidence at the $\alpha = 0.01$ level that $\beta_{age} \neq 0$ and/or $\beta_{height} \neq 0$

There is evidence of a linear association between annual salary and age and/or height (here, $p < 0.001$ as calculated using software program).

MLR Inference t-test

If the overall model is significant, then the significance could be attributed to any one of the independent variables.

Perform testing on each individual parameter to identify the relative contribution of each independent variable.

In order to test each if $\beta_i = 0$ after controlling for the other independent variables in the model, we use a t statistic:

$$t = \frac{\hat{\beta}_i}{SE_{\hat{\beta}_i}}$$

where

$SE_{\hat{\beta}_i}$ the standard error of the estimate of (in the regression model with the other independent variables included) which follows a t-distribution with n-k-1 degrees of freedom under H_0 .

MLR Inference t-test

The decision rule for a two-sided level α test is:

- ▷ **Reject $H_0 : \beta_i = 0$ if $|t_i| \geq t_{n-k-1, \alpha/2}$**
- ▷ Otherwise, do not reject $H_0 : \beta_i = 0$

where

$$t_{n-k-1, \alpha/2}$$

is the value from the t-distribution with $n - k - 1$ degrees of freedom and associated with a right hand tail probability of $\alpha/2$.

MLR Inference t-test Confidence Interval

We can calculate the two-sided $100\%(1 - \alpha)$ confidence interval for using the following formula:

$$\hat{\beta}_i \pm t_{n-k-1, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_i}$$

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We can say with $100\% \times (1 - \alpha)$ confidence that the true value of is between

$$\hat{\beta}_i - t_{n-k-1, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_i} \text{ and } \hat{\beta}_i + t_{n-k-1, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_i}$$

after controlling for the other independent variables in the model.

An Example: t-test for MLR

Is age a significant predictors of annual salary after controlling for height?

Perform a t-test at the $\alpha = 0.01$ level and calculate the 99% confidence interval for β_{age} .

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1. Set up the hypotheses and select the alpha level

$H_0 : \beta_{age} = 0$ (after controlling for height)

$H_1 : \beta_{age} \neq 0$ (after controlling for height)

$\alpha = 0.01$

2. Select the appropriate test statistic

$$t = \frac{\hat{\beta}_i}{SE_{\hat{\beta}_i}}$$

with $df = n - 3 = 100 - 3 = 97$ degrees of freedom

An Example: t-test for MLR

```
> summary(m)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.907e+02   1.884e-03  101223   <2e-16 ***
age          2.503e+00   9.862e-06  253808   <2e-16 ***
height       2.507e+00   2.541e-05   98679   <2e-16 ***
```

3. State the decision rule Determine the appropriate value from the t-distribution with 97 degrees of freedom and associated with a right hand tail probability of

$$\alpha/2 = 0.01/2 = 0.005$$

$$t_{n-k-1, \alpha/2} = t_{97, 0.005} = 2.63$$

```
> qt(0.995, df=97)
```

Decision Rule: Reject H_0 if $|t| \geq 2.63$

Otherwise, do not reject H_0

An Example: t-test for MLR

4. Compute the test statistic Using R function `summary(m)`, we get the table:

$$t = \frac{\hat{\beta}_{age}}{SE_{\hat{\beta}_{age}}} = \frac{2.503}{9.862e-06} \approx 253808 \text{ with } df = 97$$

Reject H_0 since $253808 \geq 2.627$. We have significant evidence at the $\alpha = 0.01$ level that $\beta_{age} \neq 0$ after controlling for height. We can calculate the confidence interval.

$$\hat{\beta}_{age} \pm t_{n-k-1, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_{age}} = 2.503 \pm 2.627 \cdot 9.862e-06 = (2.502970, 2.503022)$$

Computation in R:

```
> m <- lm(salary1 ~ age + height)
> confint(m, level = 0.99)
              0.5 %      99.5 %
(Intercept) 190.692341 190.702241
age          2.502970   2.503022
height       2.506966   2.507099
```